

192.168.10.0 /24 for starting

A) 31 hosts

$$2^6 - 2 = 62 \text{ host}$$

first ip: 192.168.10.0

192.168.10.00111110 → 62

192.168.10.00000000

 .001000000



192.168.10.0/26 netmask

192.168.10.63/26 broadcast

B) 10 hosts

$$2^4 - 2 = 14 \text{ hosts}$$

first ip: 192.168.10.64

192.168.10.01000000

192.168.10.01001110

 .010010000



192.168.10.64/28 netmask

192.168.10.79/28 broadcast

C) 10 hosts

$$2^4 - 2 = 14 \text{ hosts}$$

first ip: 192.168.10.80

192.168.10.01010000

 .01011110

128 → net mask

.95 → broadcast

D) 2 hosts

$$2^1 - 2 = 2 \text{ hosts}$$

192.168.10.96
 10.99 ↓

130 → net mask

99 → broadcast

Router baglantis

1.0.0.0 /30 → 2 hosts

1.0.0.3/30 → broadcast

1.0.0.4/30 →

1.0.0.7/30 → broadcast

Router 0 Configuration:

```
en
conf t
int fa 0/0
no shutdown
end
int fa 0/1
ip address 1.0.0.1 255.255.255.252
no shutdown
end
int fa 0/10.10
encapsulation dot1q 10
ip address 192.168.10.1 255.255.255.192
no shutdown
end
int fa 0/10.20
encapsulation dot1q 20
ip address 192.168.10.65 255.255.255.240
no shutdown
end
conf t
```

OSPF configuration Router 0

```
router ospf 1
router-id 1.1.1.1
network 192.168.10.0 0.0.0.63 area 10
network 192.168.10.64 0.0.0.15 area 10
network 1.0.0.0 0.0.0.3 area 0
end
wr
show ip ospf nei
```

OSPF configuration Router 1:

```
router ospf 1
router-id 2.2.2.2
network 192.168.10.0 0.0.0.63 area 0
network 192.168.10.64 0.0.0.15 area 0
network 1.0.0.0 0.0.0.3 area 0
network 1.0.0.0 0.0.0.4 area 0
show ip ospf nei
```

OSPF configuration Router 2:

```
router ospf 1
router-id 3.3.3.3
network 1.0.0.4 0.0.0.3 area 0
network 192.168.10.80 0.0.0.15 area 20
network 192.168.10.96 0.0.0.3 area 20
```

DHCP pool for vlan 40

```
ip dhcp excluded-address 192.168.10.0 192.168.10.95
ip dhcp pool for-vlan40
network 192.168.10.96 255.255.255.252
default-router 192.168.10.97
domain-name vlan40
dns-server 8.8.8.8
exit
```

NAT and ACL for vlan40

```
access-list 1 permit 192.168.10.96 0.0.0.3  
int fa0/1.10  
ip nat inside  
ex  
int fa0/10  
ip nat outside  
ex  
ip nat inside source list 1 int fa0/1 overload
```